

# SANJAY SUTHRAYE



SENIOR DATA SCIENTIST | AI & ML ENGINEER [AVAILABLE IMMEDIATELY]

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[🌐 Sanjay-Suthraye](#) | [🌐 Portfolio](#) | [📍 Kreuznacher Str. 6, 14197 Berlin, Germany](#)

## SUMMARY

Full Stack Data Scientist and AI Engineer with **7 years** of expertise integrating **Statistical Modelling and Generative AI**. Specialist in developing comprehensive intelligence systems, encompassing the design of Agentic Frameworks (**LangGraph, MCP**) and RAG solutions, as well as the implementation of robust Predictive Models (**XGBoost, Random Forest**) and **Time-Series Forecasts**. Demonstrated success at AstraZeneca and Tesco in transforming traditional analytics into scalable, containerised microservices (**Docker, FastAPI**) that yield quantifiable business results.

## SKILLS

|                        |                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core Competencies      | Predictive Modeling, Statistical Analysis, A/B Testing, Causal Inference, Time-Series Forecasting, Customer Segmentation, Churn Prediction, Market Mix Modelling, Pricing Models, ROI Optimization        |
| Machine Learning       | Regression (Linear/Logistic), Random Forest, XGBoost, LightGBM, CatBoost, Survival Analysis, K-Means, DBSCAN, Model Explainability (SHAP, LIME), Weak Supervision (Snorkel), Active Learning              |
| AI & GenAI Engineering | Generative AI, Agentic Workflows (LangGraph), Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), GraphRAG, Vector Databases, Large Language Models (LLMs), NLP (NER, Sentiment Analysis) |
| Languages & Libraries  | Python, R, SQL, PySpark, Scikit-learn, Pandas, NumPy, SciPy, spaCy, NLTK, FastAPI, Snowpark                                                                                                               |
| Cloud, MLOps & BI      | AWS (SageMaker, S3, ECS, Lambda), GCP Vertex AI, Docker, Kubernetes, MLflow, Apache Airflow, CI/CD, Tableau, Power BI, Snowflake, Apache Kafka, BigQuery                                                  |
| Leadership             | Technical Mentorship, Data Science Strategy, Cross-Functional Collaboration, Agile/Scrum, Stakeholder Communication, Ethical & Responsible AI                                                             |

## WORK EXPERIENCE

### ASTRAZENECA

*Domain: Pharmaceuticals/ Healthcare*

*Team: Oncology Business Unit*

#### AI Engineering Specialist

Sep'25 - Present

#### Data Scientist

Apr'23 - Sep'25

- Designed an Agentic AI Platform (**LangGraph, MCP**) to automate intricate marketing procedures, decreasing data access latency by **30%**. Authenticated agent outputs against statistical benchmarks to guarantee dependability.
- Engineered and implemented **AWS Step Functions** for **Precision Targeting**, deploying more than **5 predictive models** (both Supervised and Unsupervised) via **CI/CD**, and automating the retraining process to sustain model accuracy over time. Utilised **SHAP analysis** to determine feature selection and enhance business explainability.
- Exhibited the technical architecture of the **Patient Journey LLM system** at the **PMSA Conference** in New Delhi, receiving excellent recognition.
- Received the **Achiever Z Annual Award** (September 2024) for spearheading the Patient Journey Analytics project in the **Lead AI and Insights category**.
- Directed Time-Series Forecasting projects by comparing Statistical methods (**SARIMAX**) with **Deep Learning techniques (LSTM)**.
- Mitigated the inadequacy of "Ground Truth" data **utilising Weak Supervision** (Snorkel) and Active Learning techniques. Engineered Snowpark pipelines to provide augmented labels to classifiers, yielding a **10% enhancement in Ground Truth or Dependent labels**.

- Developed a comprehensive Experimentation Framework (**A/B Testing**) employing Hypothesis Testing and Central Composite Design to empirically assess model effects and enhance company strategy.

## TESCO

*Domain: UK Retail Industry*

*Team: Marketing Performance & Optimization*

### Senior Applied Data Scientist

Aug'21 - Apr'23

- Developed and containerised a **Churn Prediction** Microservice employing XGBoost and CatBoost, overseeing more than **200k** homes. Validated model effectiveness by rigorous **A/B testing**, yielding a **38% enhancement in retention**.
- Engineered a scalable **Product Choice Optimiser** and **Market Mix Modelling** framework utilising PySpark, managing millions of transactions. Employed **Adstock** transforms to model advertising decay and applied multi-objective optimisation to achieve substantial enhancements in **return on advertising investment**.
- Automated **High Value Customer (HVC) segmentation processes** employing K-Means++ on Hadoop clusters to efficiently categorise customers according to behavioural characteristics.
- Implemented machine learning models to identify clients with significant **growth potential inside cohorts**, enabling dynamic tracking that informed targeted marketing activities.
- **Led a hackathon-winning project** (April 2022) focused on employee wellness, improving workplace welfare.
- Recognised as **Employee of the Month** (April 2022) within the Spotlight program for contributions to data science.

## CARTESIAN CONSULTING

*Domain: Consulting*

### Senior Data Analyst

Feb'21 - Aug'21

- Developed a machine learning model utilising the **Levenshtein Algorithm** to identify analogous items, employing the Pareto principle to concentrate on the top 20% of suggestions.
- Developed the **Dealer Health Index** from inception, managing all aspects from data acquisition to visualisation in Tableau and Excel.
- Enhanced metrics for **62% of subpar dealers** using data-driven advice.
- Employed text analysis and **similarity scoring** to enhance product portfolios and improve sales efficiency.

## DXC TECHNOLOGY

*Domain: IT Services*

### Senior Analyst at Helpdesk

Mar'19 - Apr'20

### Technical Analyst (Contractor)

Jun'18 - Mar'19

- Developed a machine learning model utilising **named-entity recognition** to evaluate **customer reviews** and discern primary sentiment influencers.
- Enhanced team traffic procedures by regression analysis, **reducing average handling time by 15%**.
- Aligned helpdesk operations with industry standards via data-driven workflow enhancements.
- **Administered SQL/MS SQL databases** and upheld a centralised knowledge repository that enhanced information dissemination.
- Examined **call flow data** to discern trends that enhanced first-call resolution rates.

## EDUCATION

### University of Hyderabad

Jan'21 - Dec'22

Post Graduate Diploma in **Artificial Intelligence and Machine Learning**

GPA: 3.4/4.0

### AMC College of Engineering

Jan'14 - Dec'18

Bachelor of Engineering

CGPA: 7.0/10.0

## LANGUAGES

**English** - Fluent

**German** - A2 (Enrolled - Intensive Course)